Class Activity

Student Grade Manager

Time: 30–40 minutes
Work: Individual
Submission: No written submission required. **Show the completed running program to the instructor to receive the grade.**

Create a C# console application that manages a small list of students and their grades.

Your program must include a Student class with:

```
public int Id { get; set; }
public string? Name { get; set; }
public int Grade { get; set; }
```

Add a method:

```
public void Display()
{
    // Your code
}
```

The method should display the student information in this format:

```
101 - Alice - Grade: 85
```

If the student's name is null, display:

```
101 - [Name unavailable] - Grade: 85
```

In Program.cs, create at least **five students** and store them in:

```
List<Student> students = [];
```

At least one student must have a null name.

Your program must then:

```
1. Display all students.

2. Calculate and display the average grade.

3. Display the student with the highest grade.

4. Display only students whose grade is 70 or higher.
```

5. Change the name of one student through a second reference to the same Student object.

6. Display the original variable again to demonstrate that the object was modified.

7. Create two integer variables where the second is assigned from the first.

8. Change the second integer and display both values to demonstrate value-type behavior.

9. Box one student's grade into an object.

10. Unbox it back into an int and display the result.

For example, part of the program could contain:

```
Student first = students[0];
Student second = first;

second.Name = "Updated Name";

Console.WriteLine(first.Name);
```

and:

```
int grade1 = 80;
int grade2 = grade1;

grade2 = 95;

Console.WriteLine($"grade1 = {grade1}");
Console.WriteLine($"grade2 = {grade2}");
```

Your final output should be clearly organized, for example:

```
--- ALL STUDENTS ---
101 - Alice - Grade: 85
102 - Bob - Grade: 72
103 - Charlie - Grade: 91
104 - [Name unavailable] - Grade: 64
105 - Diana - Grade: 88

--- CLASS STATISTICS ---
Average Grade: 80.0
Highest Grade: Charlie - 91
```

```
--- PASSING STUDENTS ---
Alice
Bob
Charlie
Diana

--- REFERENCE TYPE TEST ---
Updated Name

--- VALUE TYPE TEST ---
grade1 = 80
grade2 = 95

--- BOXING / UNBOXING ---
Boxed Grade: 85
Restored Grade: 85
```

Grading: When finished, **run the complete program and show the working result to the instructor.** No report or submission on Moodle is required.